

RESEARCH

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Source: arxiv.org/abs/2504.03553

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Current Agent Training Uses "Flood Irrigation"

Core Issue: Indiscriminate knowledge injection leads to fragile planning, pattern collapse, and high inference costs.

Three Flawed Paradigms

- **Direct Trajectory Imitation (e.g., ReAct)**
Brittle execution, prone to pattern collapse on unexpected inputs.
- **Trial-and-Error Refinement (e.g., Reflexion)**
Inefficient and costly process.
- **Knowledge-Augmented (KnowAgent, WKM)**
Unneeded knowledge hurts performance and causes unnecessary high inference costs.

The Missing Piece: Self-Awareness

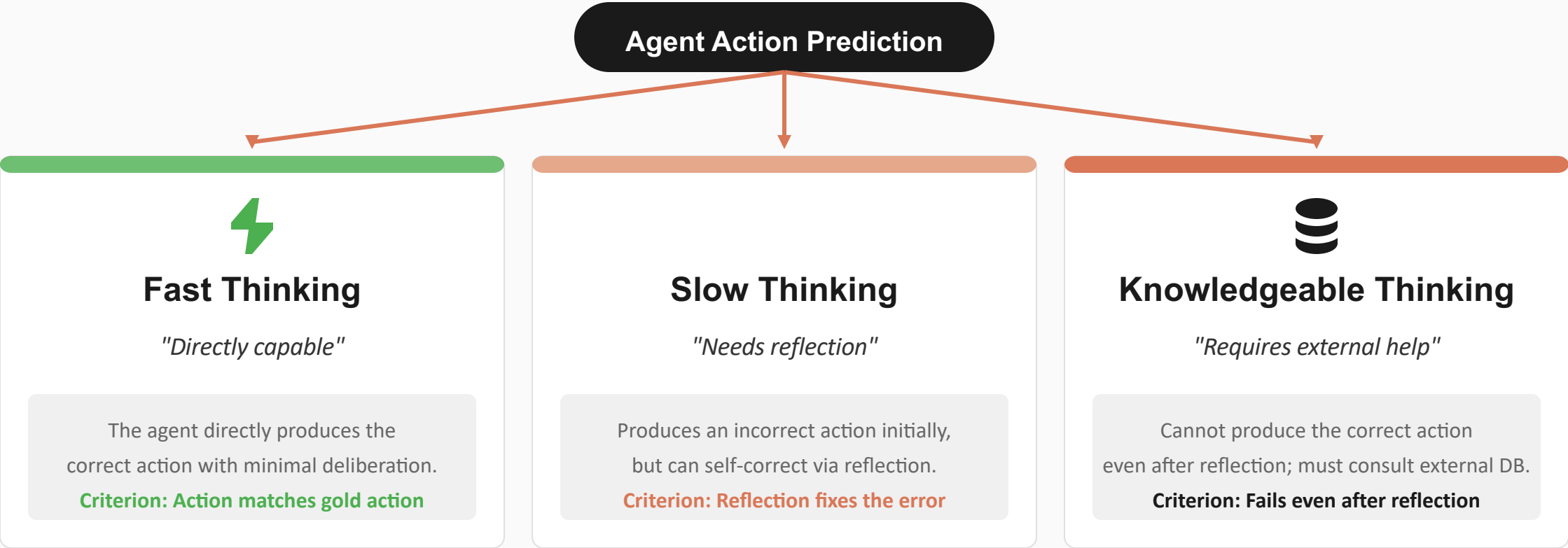
Humans possess **situational self-awareness**:

- Assess if they can handle a situation directly
- Recognize when they need to pause and reflect
- Know when to seek external help

KnowSelf aims to bring this metacognitive capability to LLM agents.

Three Situational Modes: Fast, Slow, and Knowledgeable Thinking

KnowSelf classifies the cognitive situation based on the agent's ability at each decision step.



KnowSelf Framework: Data Construction → Training → Inference

1 Self-Awareness Data

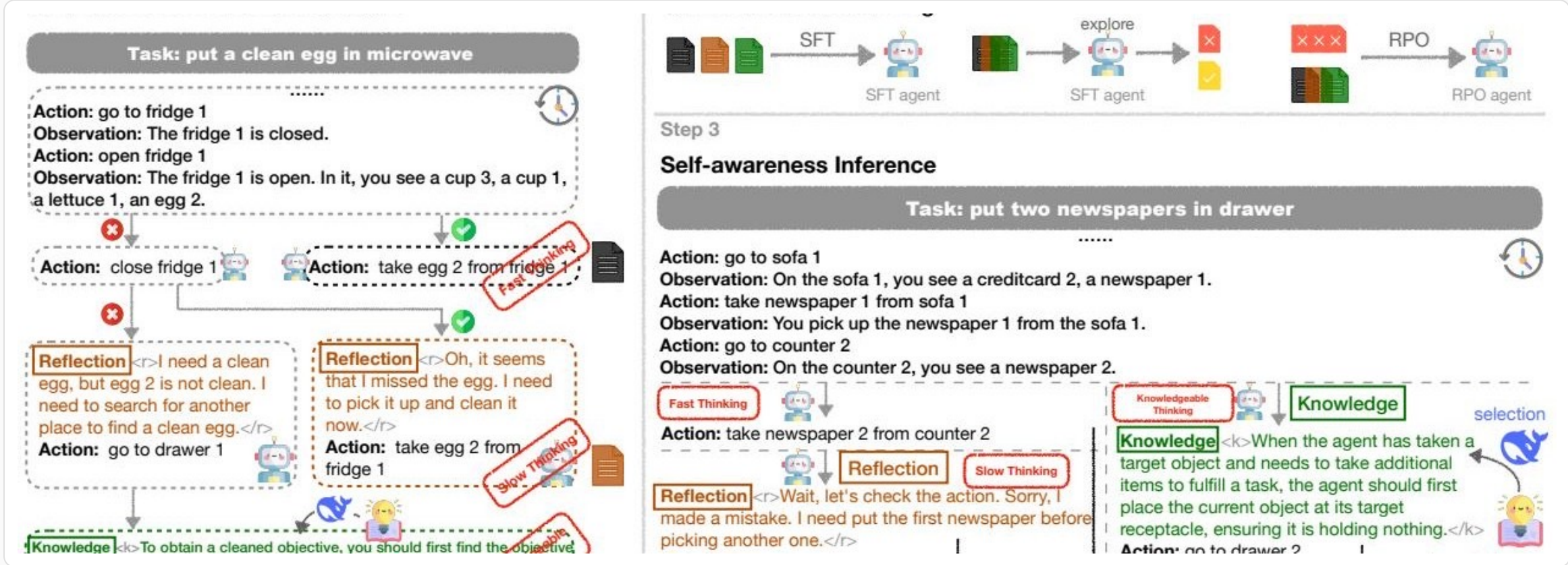
Heuristic situation judgment marks special tokens on self-explored trajectories.

2 Two-Stage Training

Supervised Fine-Tuning (SFT) followed by Reward-based Policy Optimization (RPO).

3 Self-Aware Inference

Selective token generation triggers reflection or knowledge retrieval.



KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Qualitative Proof: KnowSelf triggers reflection when making a mistake and seeks knowledge when stuck.

Task 2: "Put a saltshaker in cabinet"

DeepSeek-R1: Gets stuck trying to close the microwave.

KnowSelf: Applies Reflection to replace object directly.

(a) Task: put a hot mug in cabinet	
OpenAI-O1 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge]<k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k> Action: take egg 1 from microwave 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection]<r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r> Action: heat mug 1 with microwave 1
(b) Task: put a saltshaker in drawer	
DeepSeek-R1 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection]<r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge]<k>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</k> Action: put saltshaker 1 in/on drawer 1

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Data Evidence: KnowSelf outperforms 100%-knowledge baselines across both Llama-8B and Gemma-2B.

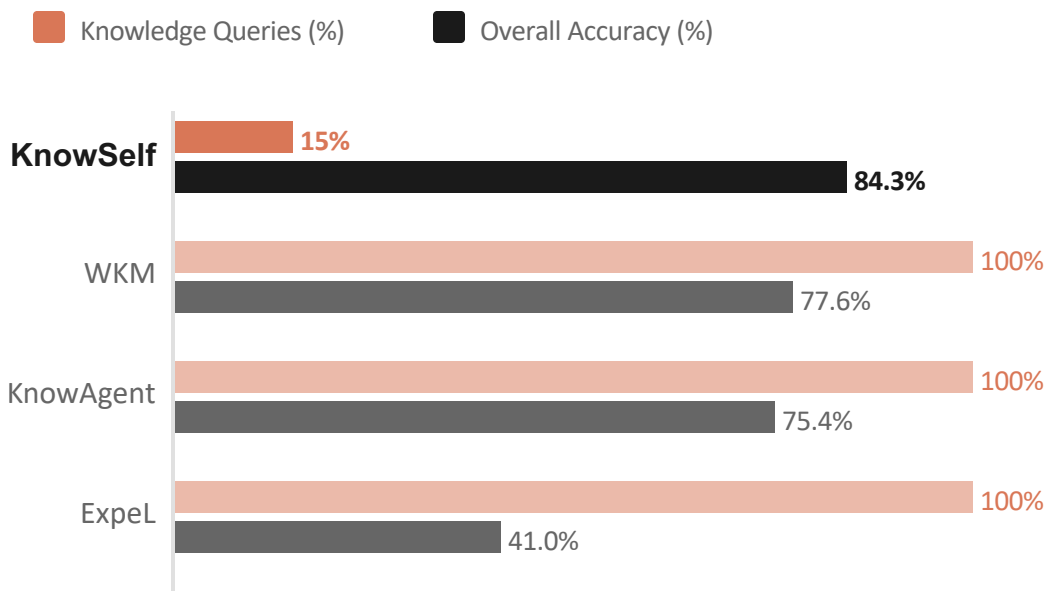
Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	Reflexion	0%							85.07
GPT-4o	ExpeL	100%							79.85
Llama-8B	Reflexion	0%							51.49
Llama-8B	ExpeL	100%							41.04
Llama-8B	KnowAgent	100%							75.37
Llama-8B	WKM	100%							77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ETO	0%							71.64
Gemma-2B	KnowAgent	100%							73.88
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Source: Extracted from ALFWorld Main Results. Highlighted rows represent KnowSelf's performance showing the efficiency of selective external knowledge usage.

Minimal Knowledge, Maximum Impact: Efficiency Over Full-Knowledge Methods

KnowSelf achieves higher accuracy while dramatically reducing knowledge dependency.

Llama-8B: Knowledge Use vs Overall Accuracy



✓ Key Efficiency Insights

~15% Knowledge usage compared to 100% in baselines.
Avoids hallucination and distraction from unneeded docs.

84.33% Best overall score on ALFWorld.
Outperforms strong models despite selective retrieval.

Scaling Effectiveness:
Gemma-2B reaches 79.85%, matching GPT-4o ExpeL.

Self-Awareness Is the Missing Piece in Agent Planning

- 1 Indiscriminate knowledge injection is wasteful and suboptimal.
- 2 Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling.
- 3 KnowSelf outperforms strong baselines (84.33% vs 76.12% for GPT-4o ReAct).
- 4 Only 15-26% knowledge usage needed versus 100% for competing methods.
- 5 The paradigm generalizes across model scales (Gemma-2B matches GPT-4o Expel).